

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: REPRODUCTION IN PLANTS (SUB-STRAND 2.1)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why do some farm plants produce offspring without seeds while others require seeds and a partner tree to make fruit? Use your notebook, your labelled dissection diagrams, your fruit classification table and the class charts as evidence. Write your answers in full sentences in the space under each prompt. In Sections 2 to 4 use the Claim-Evidence-Reasoning (CER) structure. Use examples from our unit, such as cassava, sugarcane and sweet potato in Kisumu, Kisii and Siaya, and maize, beans, pawpaw, avocado and passion fruit. Draw neat, labelled diagrams where asked.

Part 1: Making Seedless New Plants

A farmer in Kisii plants cassava stem cuttings and a woman in Siaya plants sweet potato vines. Neither uses seeds. (a) Name three natural vegetative structures we examined in class (for example from sweet potato, onion, ginger, grass or bryophyllum) and say where on each a new plant can start. (b) Explain how a cassava cutting can become a whole plant without a flower, using the idea of cell division. (c) Describe how the new plants compare with the parent plant. (d) Give one advantage and one disadvantage of growing crops this way.

Write your answer here:

Part 2: Flowers, Pollination and the Lone Tree (CER)

A compound in Nairobi has a single pawpaw tree that is covered in flowers every year but never sets fruit. Write a CER explanation. **Claim:** state why the tree does not fruit. **Evidence:** use at least two observations from our flower dissections, wire/bead model stations or animation work. **Reasoning:** explain how the features of flowers and trees that we studied at the model stations stop fruit from forming on a lone tree, and why a partner tree helps. Remember to separate pollination from fertilisation.

Write your answer here:

Part 3: Double Fertilisation and Fruit Development

(a) Describe, in order, what happens from the time compatible pollen lands on a stigma until a seed begins to form. Name the structures the pollen tube passes through. **(b)** State what each of the two fusions produces and what each product does for the seed. **(c)** Say what the ovules and the ovary become after fertilisation. **(d)** Using a fruit you sectioned in class (tomato, orange, bean pod, passion fruit or cucumber), name its placentation type and explain how its seeds show the link to fertilised ovules.

Write your answer here:

Part 4: Comparing Sexual and Asexual Reproduction in Kenyan Farming (CER)

A farmer asks you whether it is better to plant cassava from cuttings or maize from seed, and what would happen if a new disease arrived in each field. Write a CER response. Claim: state the trade-off between the two strategies. Evidence: use at least two case examples from our case cards or class work. Reasoning: link the number of parents, the type of cell division or fertilisation, and the amount of variation to survival in changing conditions. Finish by explaining why neither strategy is always best.

Write your answer here:

Part 5: Revised Model and Annotated Diagram

Look back at the first model you drew in Lesson 1, comparing a sweet potato making a new vine with a bean seed making a plant. (a) Draw a revised annotated diagram that explains how a cassava cutting, a maize or bean plant and a lone pawpaw tree each end up with, or without, new plants or fruit. Label the main structures and processes. (b) Write three to four sentences describing how your thinking changed and one question you still have.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of the phenomenon	Gives a clear, accurate claim that answers the driving question for both cassava-type and pawpaw/avocado-type plants, using Kenyan examples from the unit.	Gives a mostly accurate claim that covers both cases, with limited use of examples.	Claim is unclear, covers only one case, or contains misconceptions, for example that flowers alone guarantee fruit.
Evidence from investigations	Uses at least two specific observations from dissections, stations, animations, fruit sections or case cards, accurately and with detail.	Uses one or two observations, with some detail missing.	Gives little or no evidence, or evidence that is vague or not from our lessons.
Scientific reasoning and terminology	Correctly links mitosis and cloning, barriers to self-pollination, pollination versus fertilisation, double fertilisation (zygote and triploid endosperm), and ovule-to-seed and ovary-to-fruit development.	Links most ideas correctly, with minor errors or missing links.	Reasoning is incomplete, or confuses pollination with fertilisation or the products of the two fusions.
Comparison of sexual and asexual reproduction	Explains advantages and disadvantages of both strategies, including variation and disease risk, and why the best choice depends on the environment.	Names advantages and disadvantages of both strategies with limited explanation.	Describes only one strategy or lists points without explanation.
Annotated diagram and presentation	The diagram is neat and accurate, with all key structures and processes labelled for both pathways, and the written reflection is clear.	The diagram has most labels correct, with a few omissions, and the reflection is brief.	The diagram is missing labels or has errors, or the reflection is missing.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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